MultiThreadArray.java

Multi-threaded array generation and sorting demo

Source Code

```
import java.util.*;

class RandomArray extends Thread {
    static int[] arr = new int[10];

    public void run() {
        Random r = new Random();

        System.out.println("Random Array:");
        for (int i = 0; i < arr.length; i++) {
            arr[i] = r.nextInt(100);
            System.out.print(arr[i] + " ");
        }
        System.out.println();
    }
}

class Ascending extends Thread {
    public void run() {
        Arrays.sort(RandomArray.arr);

        System.out.println("Ascending Order:");
        for (int i : RandomArray.arr) {
            System.out.print(i + " ");
        }
        System.out.println();
    }
}

class Descending extends Thread {
    public void run() {
        System.out.println("Descending Order:");

        for (int i = RandomArray.arr.length - 1; i >= 0; i--) {
            System.out.print(RandomArray.arr[i] + " ");
        }
        System.out.println();
    }
}

public class MultiThreadArray {
    public static void main(String[] args) throws Exception {

        RandomArray t1 = new RandomArray();
        Ascending t2 = new Ascending();
        Descending t3 = new Descending();

        t1.start();
        t1.join();

        t2.start();
        t2.join();
        t3.start();
```

```
        t3.join();
    }
}
```

Output

```
Random Array:
45 12 89 23 67 10 34 78 56 91

Ascending Order:
10 12 23 34 45 56 67 78 89 91

Descending Order:
91 89 78 67 56 45 34 23 12 10
```